Payload-Aware Open-Set Interference Recognition for GEO Transponders: Literature Review

The claimed gap largely holds. I found no 2018–2026 paper that trains or evaluates a deep-learning interference detector or classifier for GEO/GSO satellites on data that models TWTA nonlinearity, IMUX/OMUX filtering, multicarrier intermodulation, dual polarization or carrier-plan metadata. Open-set rejection of unknown interference classes has been published only for generic anti-jamming, radar and GNSS signals, never for a GEO transponder. Work that partly narrows the gap (Eutelsat/CTTC/ESA, Fudan's Agon, Fraunhofer's GNSS work, open-set jamming papers) must be cited, and C1–C3 should be framed as transferring known pieces into a new setting, not as firsts in every respect.

TL;DR

- **Verdict: the gap holds, with caveats.** The GEO DL interference papers I found are from SnT/Luxembourg (Daoud et al. ICASSP 2023; Saifaldawla et al. PIMRC 2023, ICC-W 2024, IEEE TMLCN 2024), Fudan (Agon, arXiv 2026), CTTC/Eutelsat (SPAWC 2019, IEEE Commun. Mag. 2021) and KTH (Pellaco et al. ICSSC 2019). All use linear signal superposition with single-polarization inputs. They are binary detectors, either reconstruction-error or closed-set. None models the transponder chain, cross-pol, carrier plans or open-set rejection. Even the DVB-S2X reference channel guidelines treat cross-polar and co-channel interference as negligible in their multicarrier scenario.
- **The pieces exist, but in separate fields.** Payload models are mature but used only for predistortion and equalization: the Saleh TWTA, the DVB-S2 IMUX/TWTA/OMUX chain (Casini, De Gaudenzi & Ginesi 2004), and multicarrier intermodulation analysis. Open-set methods are mature but untested on satellite interference: MSP, ODIN, energy, Mahalanobis, OpenMax, OSCR, and Vaze et al.'s "good closed-set classifier" result. The most rigorous real-data, domain-shift interference ML comes from GNSS (Fraunhofer IIS DARCY/DARCII datasets and papers).
- **What to do.** Frame the novelty as "first payload-realistic, plan-conditioned, dual-polarization open-set benchmark plus a quantified payload gap." Add a carrier-cancellation or cyclic baseline for carrier-under-carrier, which a PSD cannot see. Report OSCR alongside AUROC. Validate realism against OpenDPD and the SnT data rather than claiming real-world performance.

Key Findings

1. **No GEO interference-recognition paper models the transponder.**
   - Daoud et al. (ICASSP 2023): a GSO uplink with BPSK signals, rectangular pulses and AWGN, 5 subbands, 2,500 samples per class, and 6 closed-set classes. uni
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   - Saifaldawla et al. (PIMRC 2023, ICC-W 2024, TMLCN 2024): a downlink NGSO→GSO ground-station link budget with linear superposition. TrID beats the energy detector by up to 31.23% in detection accuracy, and time-domain inputs beat frequency-domain ones. uni
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   - Agon (Fudan, arXiv 2606.14147): also a linear superposition model. It reaches AUC 0.9327 on the public NGSO-GSO set versus 0.8318 for TrID-like baselines. arxiv
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2. **Operator-motivated work names the right classes but does not model them.** Vázquez et al. (CTTC/UPC/Eutelsat/GMV/Reply; 2019 newsletter, arXiv 2007.10110) write that a mispointed antenna (cross-polarization or adjacent satellite) and misconfigured equipment (noise, intermodulation) "cover 70-80% of all interference cases". Carrier overlaps are listed separately and are not part of that 70–80%. The Commun. Mag. 2021 paper (CTTC, Eutelsat, ESA ARTES) still uses a generic autoencoder detector. That detector, as arXiv 2007.10110 describes it, "throws an alarm if the deviation trespasses a fixed threshold." The paper reports a 44% reduction in false detection probability on "real data."
3. **Open-set is the most clearly unoccupied axis for satellite interference.** Open-set jamming recognition exists in general wireless and radar work: convolutional prototype learning (Appl. Sci. 2022), OpenMax radar jamming (Remote Sens. 2022), compound-jamming EVT (IET RSN 2024), ArcCider (IET Commun. 2026).
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Open-set AMC and RF fingerprinting are also active. None of it covers satellite transponders. A 2021 interference-suppression survey (arXiv 2112.08988) says OSR/OOD has received too little attention in DL interference work. arxiv

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4. **Industry statistics support the open-set motivation.** These are industry and trade-press figures, not peer-reviewed statistics.
   - **Unknown sources.** SUIRG's early database (3,650 incidents) attributed 62% of cases to human error, cross-polarization and equipment problems. SUIRG president Robert Ames told Via Satellite (1 Aug 2005) that "another 34 percent of interference incidents are caused by unknown sources, usually because the problems appear and disappear before the operators have a chance to document it."
   - **Cross-pol and unauthorized carriers.** In Microwave Journal (14 Aug 2012), sIRG's Martin Coleman reported that "according to Intelsat, as much as 33 percent of interference is due to cross polarization leakage." The same article says Intelsat attributes 47% to unauthorized carriers.
   - **VSATs.** IRG Executive Director Martin Coleman (Dec 2014) said VSAT systems "cause approximately 40% of all interference, and are responsible for 50% of downtime." The often-cited "50–70%" figure comes from Kratos marketing copy.
   - **Frequency.** A Satellite Interference Reduction Group survey found that 93% of operators suffer interference at least once a year, more than half at least monthly, and 17% continuously.
5. **Carrier-under-carrier is a hard identifiability case.** The DVB-CID metacarrier sits at −27.5 dB relative to the host's spectral density. satellitetoday

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Commercial carrier-under-carrier detectors (R&S GSACSM, Integrasys VectorSat, Kratos Monics, Calian Decimator) work by cancelling the host carrier, not by inspecting the PSD. rohde-schwarz

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A spectrogram-only model therefore cannot be expected to identify carrier-under-carrier at realistic C/I. The cyclic branch, or a cancellation residual, is needed in practice, and the identifiability analysis (C2) should say so explicitly.
6. **GNSS is the methodological template for evaluation, not for the physics.** Fraunhofer IIS work evaluates under real-world data discrepancies, novel-class few-shot learning, MC-dropout uncertainty and controlled-versus-highway shift, with CC-BY/CC-BY-SA datasets. But GNSS interference sits far above a below-noise signal, whereas a GEO operator sees structured multicarrier spectra.

Details: Thematic Review

1. Interference detection/classification for satellite communications

Classical on-board detection (SnT, 2016–2018). Politis, Maleki, Tsinos, Chatzinotas and Ottersten proposed on-board spectrum monitoring units that decode and cancel the desired signal, then apply an energy detector to the residual. They analyzed imperfect cancellation (SPAWC 2016) and weak interference (ICASSP 2017), and used DVB-S2X pilot fields. uni
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Pérez-Neira et al. (arXiv 1802.03958) placed this within HTS signal processing and noted that on-board detection avoids the downlink noise and *transponder distortion* that degrade on-ground detection. arxiv
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This is an early acknowledgement that the payload chain corrupts ground-based observations, which supports C3.

Early DL (2019). Two papers started the DL line:

- Henarejos, Vázquez and Pérez-Neira (SPAWC 2019) used a sparse autoencoder on real Ku-band DVB-S2 captures for detection, and an LSTM to classify terrestrial interferers (LTE/UMTS/GSM) added digitally. arxiv
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- Pellaco, Singh and Jaldén (ICSSC 2019, KTH with SSC) used an LSTM to predict spectra and flagged deviations, trained on real received spectra. arxiv
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Operations (2019–2021). Vázquez et al. (CTTC, Eutelsat, Data Reply, Rheatech, GMV; ESA ARTES) brought ML into operator workflows: interference detection, flexible payload configuration and congestion prediction. acm
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SnT SmartSpace line (2023–2026). The FNR SmartSpace project (Lagunas, Ortiz, Chatzinotas) produced the on-board CNN study (Daoud et al.), the NGSO→GSO autoencoder/CAE/VAE/TrID line (Saifaldawla et al.) and public datasets.
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All are binary or subband-localization tasks with a single carrier per scenario, and none models the payload.

Recent Chinese and other work. Agon (Fudan, 2026) adds MAE pre-training of a dual-attention transformer, HOS-augmented attention, wavelet regularization and a multi-task head, and runs cross-dataset/OOD robustness tests. arxiv
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This is covariate shift, not unknown-class rejection. Several closed-set satellite interference classifiers cover single-tone, narrowband, broadband noise, pulse, sweep and multi-tone interference in AWGN (ACM 2018; IEEE conference 2024; ICECAI 2025).
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A Kernel-FastICA anti-jamming paper explicitly says existing methods “lack effective modeling of the physical distortion mechanisms caused by transponder nonlinearities,” which is direct support for the gap claim, although that paper does suppression rather than recognition. nih
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On-board vs. ground. On-board approaches (Politis, Daoud, the AI-accelerator review in *Aerospace* 2023) observe the uplink before the TWTA. The project's ground-downlink view is the one in which payload IM and external interference become confounded.

2. GNSS interference/jamming classification with DL

Fraunhofer IIS (the DARCY and DARCII projects) is the most prolific group:

- Brieger et al. (ION GNSS+ 2022) fuse spectrum and time-domain data and use MC-dropout entropy as a reliability measure. researchgate
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- Ott et al. (ICL-GNSS 2024) apply few-shot quadruplet learning.
- Heublein et al. (ION GNSS+ 2024, arXiv 2503.23775) evaluate supervised and unsupervised methods under real-world data discrepancies. arxiv
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- Further work covers VAE disentanglement (arXiv 2504.10556), LLM-based characterization (arXiv 2501.05079), attention fusion with AoA (arXiv 2507.14167), and the Sionna-simulated S-ICDF dataset (arXiv 2607.03411).

  Datasets include the Highway 1 dataset (IEEE DataPort, doi:10.21227/xpm9-tt28, CC BY 4.0), Highway 2, LC-laboratory and Jammertest 2025 (Zenodo, CC BY-SA 4.0).
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  Other groups contribute too: Mehr and Dovis (IEEE TAES) on ResNet-based classification from terrestrial and LEO stations, and an MDPI *Electronics* 2025 bit-depth study. doi
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  What transfers to GEO: condition-based splits (location, sensor, day), explicit "novel class" protocols, uncertainty-based rejection, and spectrogram-plus-feature fusion. **What does not:** the GNSS interference catalogue (chirp jammers, DME pulses), which matches the held-out unknowns rather than the GEO known classes, and the below-noise desired-signal regime.

3. Generic RF/spectrum DL

- TorchSig's Sig53 (5M samples, 53 classes) and WidebandSig53 (550k samples, ~2M signals; Boegner et al., arXiv 2211.10335) move to wideband detection and recognition on complex spectrograms with CNN and transformer detectors. They are good pre-training and augmentation sources, but contain no satellite payload effects or polarization. arxiv
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- Tekbıyık et al. (arXiv 2003.08359) showed that CNNs on spectral-correlation-function inputs beat raw-signal networks on real cellular captures. arxiv
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  This is direct support for the cyclic branch.
- Cyclo-AMC (FPT 2025) reports that cyclostationary features improve DNN generalization. phwl
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- “Keep It Simple” (arXiv 2303.03326) found that small CNNs suffice for interference classification. This supports including a small-model baseline and reporting FLOPs and latency.

4. Open-set recognition and OOD detection

Foundations: OpenMax (Bendale & Boult, CVPR 2016); MSP baseline (Hendrycks & Gimpel, ICLR 2017); ODIN (Liang et al., ICLR 2018); Mahalanobis (Lee et al., NeurIPS 2018); energy score (Liu et al., NeurIPS 2020); OSCR and “agnostophobia” (Dhamija et al., NeurIPS 2018); ARPL (Chen et al., TPAMI 2021); and “Open-Set Recognition: a Good Closed-Set Classifier is All You Need” (Vaze et al., ICLR 2022), which shows that MSP/MLS baselines with strong closed-set training match or beat specialized OSR methods. Surveys: Geng et al. (TPAMI 2020) and Yang et al. (generalized OOD survey).

RF applications:

- open-set AMC: DC-LSTM with center loss and Weibull (arXiv 2002.12037); CIR reconstruction (arXiv 2312.13023); FSOS-AMC (arXiv 2410.10265); adversarial attacks on AMOSR (arXiv 2405.05022) arxiv+3
- open-set RF fingerprinting (arXiv 2306.13895) arxiv
- expert-feature-augmented open-set classification (arXiv 2302.03749) arxiv
- the open-set jamming papers listed above

Pitfalls for the evaluation design:

- Results depend on which classes are held out (“semantic” vs. “nuisance” novelty). The three unknowns (pulsed, wideband noise, intermittent bursts) differ in how close they are to the known classes. Wideband noise resembles overdrive-IM spectral regrowth, so report per-unknown AUROC.
- Vaze et al. imply that closed-set accuracy drives OSR. Report closed-set accuracy jointly, through OSCR.
- AUROC hides operating-point behaviour. Add FPR@95TPR.
- Tuning thresholds or ODIN temperatures on the test unknowns is leakage. Use a separate validation unknown, or none at all.

5. Satellite payload modeling

- **Amplifier models.** The Saleh (1981) TWTA model remains the standard AM/AM–AM/PM form. Rapp (SSPA) and the memory polynomial are the usual alternatives.
- **DVB-S2 reference channel.** The DVB-S2 standard (ETSI EN 302 307) and Casini, De Gaudenzi & Ginesi (IJSCN 2004) define the IMUX–TWTA–OMUX transponder, with conventional and linearized TWTA curves, a phase-noise mask, and static and dynamic predistortion. scispace
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  The DVB-S2X implementation guidelines (TR 102 376-2) include a multicarrier scenario with six carriers through one TWTA and an adjacent-beam model. They explicitly assume cross-polar and co-channel contributions are negligible. dvb
  C1’s dual-pol extension therefore goes beyond the reference channel and must be justified with XPD measurement literature.

- **Multicarrier distortion.** Beidas (IEEE TCOM) analyzes intermodulation in multicarrier transponders. researchgate
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  Allegue-Martínez et al. (*Electronics Letters* 2014) show that IMUX/OMUX group delay "masks" the TWTA characteristics in multicarrier DVB-S2. wiley
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  This matters for identifiability: payload IM appears through a filtered, memory-ful channel. NPR testing of TWTAs (ResearchGate 2021) gives a realism check for loading sweeps. researchgate
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- **Measured PA data.** OpenDPD (TU Delft; ISCAS 2024; v2 arXiv 2507.06849) supplies measured PA data (DPA_200MHz) and PyTorch PA models. arxiv
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  It is not a TWTA, so use it to test amplifier-model mismatch, not as GEO ground truth.

6. Interference mechanisms and operational practice

- **Cross-polarization.** Causes include feed misalignment, rain depolarization and insufficient XPD (see the dual-polarised HTS and JPL IPN references).
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- **Carrier ID.** ITU-R S.2062 (2014) and the DVB-CID metacarrier support source identification. satellitetoday
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  The FCC required CID for occasional-use video uplinks from June 2016. satellitetoday
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- **PCMA/carrier-under-carrier.** Blind PCMA detection uses SIC/ICA, and its detection limits have been analyzed information-theoretically. ieee
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- **Commercial monitoring.** Kratos Monics/SatGuard, Integrasys VectorSat ("up to 200 carriers per second"), R&S GSACSM and Calian Decimator all advertise carrier-under-carrier detection.
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  These are vendor claims and have not been independently benchmarked.
- **Intentional jamming.** In March 2026 Kratos reported sustained Iran-origin jamming affecting at least six commercial GEO satellites, including on/off cycling. kratosspace
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  This is exactly the "intermittent burst" unknown class.

7. Sim-to-real and domain shift

RF-specific evidence is thin. RadioML's over-the-air study, Heublein et al.'s real-world discrepancy evaluation, embedding-based domain generalization for RF (arXiv 2510.23186) and synthetic-to-real RF VLM tests (arXiv

2605.04676) all find degradation without calibration or adaptation. arxiv
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That finding supports C3's condition-based splits. No GEO interference study reports results under domain shift except Agon, which uses cross-constellation and cross-dataset tests.

Leakage risks to avoid: the same carrier-plan instance in train and test; the same random seed for the payload transfer function; spectrogram normalization fitted on all data.

8. Classical baselines

- **Energy detection** is the universal baseline in the SnT papers (Politis; Saifaldawla TMLCN). uni
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- **CA-CFAR** (Finn & Johnson 1968) is standard for spectral peak detection.
- **Cyclostationary detection** (Gardner 1991) is more robust at low SNR but expensive.
- **Feature-plus-gradient-boosting** (XGBoost, Chen & Guestrin 2016) is a fair "expert" baseline.

Additional baselines to add: signal-cancellation energy detection (EDSC, per Politis), which approximates operator carrier-under-carrier tools; a reconstruction autoencoder (Saifaldawla-style); and the WBSig53 detector as a pre-trained baseline.

9. Identifiability

No paper offers a formal identifiability analysis for satellite interference classes, which makes C2's analysis a real contribution. The physics-based cues are:

Payload IM vs. external carrier. IM products sit at predictable frequencies ($2f_1 - f_2$ etc.) given the carrier plan, and scale with input back-off at roughly a 3:1 dB slope for third-order products. External carriers do neither. *Needs: plan mask plus a back-off sweep.*
Cross-pol leakage vs. co-pol unplanned carrier. Leakage reproduces a carrier present on the opposite polarization at reduced level (XPD). *Needs: the cross-pol channel.*
Carrier-under-carrier. Invisible in the PSD at low C/I. It needs cyclic features at the under-carrier symbol rate, or a cancellation residual.
ASI vs. unauthorized carrier. These are spectrally similar. Separating them needs plan metadata, geolocation or CID, which are beyond the observables. Report this pair as confusable by design.

Information-theoretic analysis of blind PCMA detection is the closest existing formal treatment.

10. Candidate venues

Venue	Fit	Comparable papers
IEEE TAES	Strong for a full journal paper with payload modeling and evaluation rigor	Mehr & Dovis GNSS DNN classification
IEEE TMLCN	Strong: ML-for-comms, open-access	Saifaldawla et al. 2024 (GenAI NGSO detection)
IEEE TCCN	Good for open-set signal recognition	Open-set wireless identification in ISM band

Venue	Fit	Comparable papers
IEEE OJ-COMS	Good, fast, open access	—
IJSCN (Wiley)	Domain-native; reviewers know DVB-S2 channels	Casini et al. 2004; Bhattacharyya et al. 2023 ML survey
ICASSP / SPAWC	Short version or dataset-generator paper	Daoud 2023; Henarejos 2019; Politis 2016/17
ICC/GLOBECOM satellite workshops	Workshop version	Saifaldawla ICC-W 2024
ASMS/SPSC, ICSSC (AIAA/IET)	Operator audience	Pellaco 2019
IEEE Aerospace Conf., MILCOM	Applied/defense audience; jamming angle	—

Gap Matrix

Legend: ✓ modeled; ✕ not modeled; ◐ partial or incidental.

Paper	TWTA NL / IBO	IMUX/OMUX	Multicarrier IM	Dual-pol / XPD	Carrier plan	Open-set	Data	Met
Politis et al. SPAWC'16/ICASSP'17	✗	✗	✗	✗	✗	✗	Analytical/sim	Pd vs IS
Henarejos et al. SPAWC'19	✗	✗	✗	✗	✗	✗	Real Ku DVB-S2 + digital interferers	Accurac
Pellaco et al. ICSSC'19	✗	✗	✗	✗	✗	✗	Real spectra (operator)	Detecti rates
Vázquez et al. Commun. Mag.'21	✗	✗	✗	✗ (motivation only)	✗	✗	"Real data" (Eutelsat context)	False-detecti reducti
Daoud et al. ICASSP'23	✗	✗	✗	✗	◐ (subband grid)	✗	Sim BPSK, 6 classes	Accurac
Saifaldawla et al. PIMRC'23	✗	✗	✗	✗	✗	✗	Sim NGSO→GSO	Accurac
Saifaldawla et al. ICC-W'24	✗	✗	✗	✗	✗	✗	Sim NGSO→GSO	Pd (+11% ED)
Saifaldawla et al. TMLCN'24	✗	✗	✗	✗	✗	✗	Public NGSO-GSO set	Accurac
Agon (Yang et al.) arXiv'26	◐ (testbed only)	✗	✗	✗	✗	✗ (covariate OOD only)	NGSO-GSO + NGSO-NGSO + testbed	AUC, F1
Sat. interference CNN (ACM'18; IEEE'24; ICECAI'25)	✗	✗	✗	✗	✗	✗	Sim AWGN	Accurac
Open-set jamming (Appl. Sci.'22; IET'26)	✗	✗	✗	✗	✗	✓	Sim generic jamming	Acc., AU
Fraunhofer GNSS (Heublein'24 etc.)	✗	✗	✗	✗	✗	◐ (novel class, few-shot)	Real + controlled	F1, acc.
DVB-S2X TR 102 376-2 (reference channel)	✓	✓	✓	✗ (assumed negligible)	◐	n/a	Reference model	BER/MO

Paper	TWTA NL / IBO	IMUX/OMUX	Multicarrier IM	Dual-pol / XPD	Carrier plan	Open-set	Data	Met
This project (planned)	✓	✓	✓	✓	✓	✓	Sim + realism checks	Pd@Pfa, macro-F AUROC

Annotated Bibliography

Tags: C1 = dataset generator; C2 = plan-conditioned open-set recognizer and identifiability; C3 = payload gap / generalization; components: [SIM] simulator, [MOD] model, [OSR] open-set, [BASE] baselines, [EVAL] evaluation. DOIs marked "(verify)" are recalled, not confirmed during this search.

A. Satellite interference detection/classification

1. **Daoud, S., Eappen, G., Ortiz, F., Lagunas, E., Martins, W., Chatzinotas, S.** "CNN-Based On-Board Interference Detection in Satellite Systems: An Analysis of Dataset Impact on Performance." ICASSP 2023. Landing: https://ieeexplore.ieee.org/document/10193395/ · OA PDF: https://orbilu.uni.lu/bitstream/10993/55407/1/ICASSP_2023_Saed.pdf. A CNN on board a GSO satellite classifies which of 5 subbands contains NGSO interference, using BPSK signals with rectangular pulses, fs = 5 GHz around 29.55 GHz, and 2,500 samples per class. uni
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Reduced-size magnitude FFT gave the best accuracy. uni
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There is no payload model, and the classification is closed-set. Data are public on SmartSpace. Tags: C1 (contrast), C3; [BASE][EVAL].
2. **Saifaldawla, A., Ortiz, F., Lagunas, E., Daoud, S., Chatzinotas, S.** "NGSO-To-GSO Satellite Interference Detection Based on Autoencoder." IEEE PIMRC 2023. Landing: https://www.semanticscholar.org/paper/2e58ac0017ce6245ed8328d3e892e0fe7c7360a4 · OA PDF: paywalled (ResearchGate copy only). An autoencoder flags LEO downlink interference at a GSO terminal from reconstruction error in the time and frequency domains. It uses a single MODCOD and a single NGSO satellite. uni
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Tags: C2 (anomaly-detection contrast); [BASE].
3. **Saifaldawla, A., Ortiz, F., Lagunas, E., Chatzinotas, S.** "Convolutional Autoencoders for Non-Geostationary Satellite Interference Detection." IEEE ICC Workshops 2024, pp. 1334–1339. OA PDF: https://orbilu.uni.lu/bitstream/10993/60597/1/Moatssim__ICC__Final.pdf. Two CAE models are trained only on interference-free GSO signals and improve detection probability by up to 11% over conventional methods. uni
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Tags: [BASE].
4. **Saifaldawla, A., Ortiz, F., Lagunas, E., Adam, A. B. M., Chatzinotas, S.** "GenAI-based Models for NGSO Satellites Interference Detection." IEEE TMLCN, vol. 2, pp. 904–924, 2024. DOI: 10.1109/TMLCN.2024.3418933. OA PDF: https://orbilu.uni.lu/bitstream/10993/61484/1/TMLCN_Final.pdf (CC BY 4.0). VAE and transformer (TrID) reconstruction detectors, using real satellite positions and link budgets, are tested across QPSK, 8PSK

and 16APSK, bandwidth overlap and off-axis angles. uni
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TrID beats ED by up to 31.23%, and time-domain inputs beat spectra. uni
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This is the closest modern GEO-ground baseline. Tags: C3; [BASE][EVAL].

5. **Yang, B., Yang, C., Chen, Z., Qiu, K., Gao, Y.** “Agon: A Semi-Supervised Framework for Robust Satellite Interference Detection.” arXiv:2606.14147, 2026. Landing: https://arxiv.org/abs/2606.14147 · OA PDF: https://arxiv.org/pdf/2606.14147. Combines MAE pre-training of a dual-attention transformer, HOS attention, wavelet loss and a multi-task head. Reports AUC 0.9327 and F1 0.8351 on the public NGSO-GSO set, and 0.9085/0.9055 on its NGSO-NGSO set, with OOD robustness tests. arxiv
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Uses a linear superposition model, and the task is binary. This is the strongest recent competitor. Tags: C2, C3; [MOD][EVAL].
6. **Pellaco, L., Singh, N., Jaldén, J.** “Spectrum Prediction and Interference Detection for Satellite Communications.” ICSSC 2019 (IET). OA PDF: https://arxiv.org/pdf/1912.04716. An LSTM predicts interference-free spectra and flags deviations, trained on real received spectra (KTH with SSC). arxiv
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Tags: [BASE], realism.
7. **Henarejos, P., Vázquez, M. Á., Pérez-Neira, A. I.** “Deep Learning for Experimental Hybrid Terrestrial and Satellite Interference Management.” IEEE SPAWC 2019. DOI: 10.1109/SPAWC.2019.8815532. OA PDF: https://arxiv.org/pdf/1906.03012. Uses a sparse autoencoder on real Ku-band DVB-S2 IQ and an LSTM classifier of LTE, UMTS and GSM interferers. arxiv
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9. **Vázquez, M. Á., et al.** “On the Use of AI for Satellite Communications.” IEEE ComSoc SSC Newsletter, 2019. OA PDF: https://arxiv.org/pdf/2007.10110. States that mispointed antennas (cross-polarization or adjacent satellite) and misconfigured equipment (noise, intermodulation) “cover 70-80% of all interference cases.” It describes the autoencoder detector as one that “throws an alarm if the deviation trespasses a fixed threshold.” Tags: class taxonomy (C2).
10. **Politis, C., Maleki, S., Tsinos, C., Chatzinotas, S., Ottersten, B.** “On-board the Satellite Interference Detection with Imperfect Signal Cancellation.” IEEE SPAWC 2016. Paywalled. Energy detection after decoding and cancelling the desired signal, with analytical Pd under imperfect cancellation. mdpi
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Tags: [BASE] (EDSC).
11. **Politis, C., Maleki, S., Tsinos, C., Chatzinotas, S., Ottersten, B.** “Weak Interference Detection with Signal Cancellation in Satellite Communications.” ICASSP 2017. OA PDF: https://orbilu.uni.lu/bitstream/10993/29259/1/icassp_final.pdf. Extends EDSC to weak interference. uni
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Tags: [BASE].

12. **Pérez-Neira, A. I., Vázquez, M. Á., Maleki, S., Shankar, M. R. B., Chatzinotas, S.** “Signal Processing for High Throughput Satellite Systems: Challenges in New Interference-Limited Scenarios.” OA PDF: https://arxiv.org/pdf/1802.03958. Notes that on-ground detection is degraded by transponder distortion, and compares on-board CED, EDSCP and EDSCD. arxiv
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    Tags: C3 motivation.
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    Tags: venues, context.
15. **Fourati, F., Alouini, M.-S.** “Artificial Intelligence for Satellite Communication: A Review.” OA PDF: https://arxiv.org/pdf/2101.10899. Tags: context.
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    Tags: [EVAL] (latency).
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20. **Xiong, H.** “Research on Interference Detection and Identification Algorithms for Satellite Communication Systems.” ICECAI 2025. Landing: https://www.semanticscholar.org/paper/08c7cf481ab890ff1f6f8c45353fa4c4bc9406bc. Closed-set classification of five classes. semanticscholar
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    This is a suppression paper, not recognition. Tags: C3 gap evidence.
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channelizer subchannels. mdpi
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Tags: [BASE] (CW).

B. GNSS interference ML

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    Tags: C3; [EVAL][OSR].
24. **Brieger, T., Raichur, N. L., Jdidi, D., Ott, F., Feigl, T., van der Merwe, J. R., Rügamer, A., Felber, W.** “Multimodal Learning for Reliable Interference Classification in GNSS Signals.” ION GNSS+ 2022, pp. 3210–3234. Landing: https://www.researchgate.net/publication/364056875. arxiv
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    arxiv
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    Fuses spectrum and time data with MC-dropout entropy for reliability. researchgate
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    Tags: C2 (multimodal fusion, uncertainty).
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27. **Heublein, L., et al.** “VAE-based Feature Disentanglement for Data Augmentation and Compression in Generalized GNSS Interference Classification.” OA: https://arxiv.org/abs/2504.10556. Tags: C3.
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29. **“Real-time Pre-Correlation GNSS Interference Classification with Lightweight Learned Algorithms.”** OA PDF: https://arxiv.org/pdf/2607.28404. Tags: [EVAL] (latency).
30. **Mehr, I. E., Dovis, F.** “A Deep Neural Network Approach for Classification of GNSS Interference and Jamming.” IEEE TAES, 2024 (verify DOI). Landing: https://www.researchgate.net/publication/384100236. Covers terrestrial and LEO monitoring. researchgate
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    Tags: venue comparable.
31. **“A Machine Learning Evaluation of the Impact of Bit-Depth for the Detection and Classification of Wireless Interferences in GNSS.”** *Electronics* 14(6):1147, 2025. DOI: 10.3390/electronics14061147. OA. Tags: [EVAL] (quantization).

C. Generic RF DL

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33. **Boegner, L., Gulati, M., Vanhoy, G., et al.** “Large Scale Radio Frequency Signal Classification” (Sig53/TorchSig). OA PDF: https://arxiv.org/pdf/2207.09918. Tags: [SIM] tooling.
34. **Boegner, L., Vanhoy, G., Vallance, P., Gulati, M., Feitzinger, D., Comar, B., Miller, R. D.** “Large Scale Radio Frequency Wideband Signal Detection & Recognition.” OA PDF: https://arxiv.org/pdf/2211.10335. WBSig53 (550k samples, 53 classes) with segmentation and detection networks, including transformers, on spectrograms. arxiv
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Tags: [MOD][BASE].
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Tags: C2 cyclic branch.
37. **“Cyclo-AMC: Automatic Modulation Classification on Versal.”** FPT 2025. OA PDF: http://phwl.org/assets/papers/cycloamc_fpt25.pdf. Cyclostationary features improve DNN generalization. Tags: C2, C3.
38. **“Keep It Simple: CNN Model Complexity Studies for Interference Classification Tasks.”** OA PDF: https://arxiv.org/pdf/2303.03326. Tags: [EVAL] (params/FLOPs).
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40. **“Interference Suppression Using Deep Learning: Current Approaches and Open Challenges.”** OA: https://arxiv.org/abs/2112.08988. Flags OSR/OOD as under-studied. arxiv
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Tags: gap evidence.
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Tags: C1 (randomizing PA models).
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44. **Perez, E., Strub, F., de Vries, H., Dumoulin, V., Courville, A.** “FiLM: Visual Reasoning with a General Conditioning Layer.” AAAI 2018. OA: https://arxiv.org/abs/1709.07871. A natural mechanism for plan conditioning, as an alternative to the mask channel. Tags: C2 [MOD].

D. Open-set / OOD

45. **Bendale, A., Boult, T.** “Towards Open Set Deep Networks” (OpenMax). CVPR 2016. OA: https://arxiv.org/abs/1511.06233. Tags: [OSR].
46. **Hendrycks, D., Gimpel, K.** “A Baseline for Detecting Misclassified and Out-of-Distribution Examples.” ICLR 2017. OA: https://arxiv.org/abs/1610.02136. Tags: [OSR] (MSP).
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48. **Lee, K., Lee, K., Lee, H., Shin, J.** “A Simple Unified Framework for Detecting OOD Samples and Adversarial Attacks” (Mahalanobis). NeurIPS 2018. OA: https://arxiv.org/abs/1807.03888. Tags: [OSR].

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50. **Dhamija, A. R., Günther, M., Boult, T.** “Reducing Network Agnostophobia” (OSCR). NeurIPS 2018. OA: https://arxiv.org/abs/1811.04110. Tags: [OSR][EVAL].
51. **Vaze, S., Han, K., Vedaldi, A., Zisserman, A.** “Open-Set Recognition: a Good Closed-Set Classifier is All You Need.” ICLR 2022. OA: https://arxiv.org/abs/2110.06207. Tags: [OSR][EVAL].
52. **Chen, G., Peng, P., Wang, X., Tian, Y.** “Adversarial Reciprocal Points Learning for Open Set Recognition.” TPAMI 2021. OA: https://arxiv.org/abs/2103.00953. Tags: [OSR] (extra baseline).
53. **Geng, C., Huang, S.-J., Chen, S.** “Recent Advances in Open Set Recognition: A Survey.” TPAMI 2020. OA: https://arxiv.org/abs/1811.08581. Tags: [OSR].
54. **Yang, J., Zhou, K., Li, Y., Liu, Z.** “Generalized Out-of-Distribution Detection: A Survey.” OA: https://arxiv.org/abs/2110.11334. Tags: [OSR][EVAL].
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This is the closest OSR-interference precedent. Tags: C2 [OSR].
56. **“An Open-Set Jamming Signal Recognition Method Based on Domain-Specific Network.”** IET Commun., 2026. DOI: 10.1049/cmu2.70198. Landing: https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cmu2.70198. Tags: [OSR].
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58. **Xiao, et al.** “Open-set Recognition of Compound Jamming Signal Based on Multi-task Multi-label Learning.” IET RSN, 2024. DOI: 10.1049/rsn2.12561. Tags: [OSR] (compound events).
59. **“MDPAR: Open-Set Recognition Method for Active Radar Jamming Signals.”** Preprints.org, 2025. OA: https://www.preprints.org/manuscript/202503.2079. Not peer-reviewed. Tags: [OSR].
60. **“Open Set Modulation Recognition Based on Dual-Channel LSTM Model.”** OA PDF: https://arxiv.org/pdf/2002.12037. Tags: [OSR].
61. **“Class Information Guided Reconstruction for Automatic Modulation Open-Set Recognition.”** OA: https://arxiv.org/abs/2312.13023. Tags: [OSR].
62. **Zhang, H., Zhou, F., Wu, Q., Yuen, C.** “FSOS-AMC: Few-Shot Open-Set Learning for AMC.” OA: https://arxiv.org/abs/2410.10265. Tags: [OSR].
63. **“Adversarial Threats to Automatic Modulation Open Set Recognition in Wireless Networks.”** OA: https://arxiv.org/abs/2405.05022. Tags: [EVAL] (robustness).
64. **“Open-Set RF Fingerprinting via Improved Prototype Learning.”** OA PDF: https://arxiv.org/pdf/2306.13895. Tags: [OSR].
65. **“Open Set Wireless Signal Classification: Augmenting Deep Learning with Expert Feature Classifiers.”** OA: https://arxiv.org/abs/2302.03749. A hybrid of DL and an expert classifier, directly relevant to the GBM baseline. arxiv
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Tags: [BASE][OSR].

E. Payload modeling

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67. **Casini, E., De Gaudenzi, R., Ginesi, A.** “DVB-S2 Modem Algorithms Design and Performance over Typical Satellite Channels.” IJSCN 22(3), 2004. DOI: 10.1002/sat.791. Landing: https://onlinelibrary.wiley.com/doi/abs/10.1002/sat.791. Paywalled. Covers the IMUX–TWTA–OMUX chain,

predistortion and phase noise. wiley
N̈
Tags: C1 [SIM].
68. **Morello, A., Mignone, V.** “DVB-S2: The Second Generation Standard for Satellite Broad-band Services.” Proc. IEEE 94(1), 2006. DOI: 10.1109/JPROC.2005.861013 (verify). OA copy: https://scispace.com/pdf/dvb-s2-the-second-generation-standard-for-satellite-broad-1hexqkf3ux.pdf. Tags: C1.
69. **ETSI/DVB.** “DVB-S2X Implementation Guidelines” (TR 102 376-2; DVB BlueBook A171-2). OA PDF: https://dvb.org/wp-content/uploads/2020/02/A171-2_DVB-S2X_Implementation-Guidelines_Draft-TR-102-376-2_v121_Apr-2020.pdf. Gives conventional and linearized TWTA curves and the multicarrier scenario (6 carriers per TWTA). Cross-polar and co-channel contributions are assumed negligible. Tags: C1 traceability.
70. **Allegue-Martínez, M., et al.** “Digital Linear Pre-compensation Technique to Enhance Predistortion Performance in Multicarrier DVB-S2 Satellite Communication Systems.” *Electron. Lett.*, 2014. DOI: 10.1049/el.2014.1172. Uses a 36 MHz transponder whose IMUX/OMUX group delay masks the TWTA. wiley
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Tags: C1, C2.
71. **Beidas, B. F.** “Intermodulation Distortion in Multicarrier Satellite Systems: Analysis and Turbo Volterra Equalization.” IEEE TCOM, 2011. Landing: https://www.researchgate.net/publication/224232286. Paywalled. Tags: C1, C2 (IM identifiability).
72. **“NPR and Intermodulation Distortion Modelling of Travelling-Wave Tube Amplifier of Telecommunication Spacecraft.”** Landing: https://www.researchgate.net/publication/349494998. Tags: C1 validation (NPR).
73. **“Widely Linear Augmented Extreme Learning Machine Based Impairments Compensation for Satellite Communications.”** OA PDF: https://arxiv.org/pdf/2506.14557. Models TWTA, IMUX/OMUX and multicarrier IM for compensation. arxiv
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This shows the models are standard in satellite ML, but not in interference recognition. Tags: C1, gap.
74. **Wu, Y., Singh, G. D., Beikmirza, M., de Vreede, L. C. N., Alavi, M., Gao, C.** “OpenDPD: An Open-Source End-to-End Learning & Benchmarking Framework for Wideband PA Modeling and DPD.” ISCAS 2024. OA PDF: https://arxiv.org/pdf/2401.08318. Measured DPA_200MHz data. Reports ACPR −44.69/−44.47 dBc and EVM −35.22 dB. arxiv
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Tags: C1 realism, C3.
75. **Wu, Y., et al.** “OpenDPDv2.” OA: https://arxiv.org/abs/2507.06849. Tags: C1.
76. **Patent: “Apparatus and Method for Optimizing the Power Utilization of a Satellite Spot Beam Transponder for a Multicarrier Transmission.”** US 10103804. OA: https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/10103804. Describes power robbing, weak-signal suppression and IM versus IBO. uspto
N̈
Tags: C2 identifiability.

F. Operational practice, statistics, regulation

77. **ITU-R S.2062-0 (09/2014).** “Carrier Identification System for FSS Occasional Use Earth Station Transmissions.” OA PDF: https://www.itu.int/dms_pubrec/itu-r/rec/s/R-REC-S.2062-0-201409-I!!PDF-E.pdf. Tags: C2 (unauthorized carrier).
78. **Via Satellite (2014).** “Preventing Adjacent Satellite Interference.” https://www.satellitetoday.com/technology/2014/03/26/preventing-adjacent-satellite-interference/. Gives CID metacarrier parameters (−27.5 dB relative to host spectral density). satellitetoday
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Tags: C2.

79. **Bates, J., Via Satellite (1 Aug 2005).** “Satellite Operators Target Interference Problem.” https://www.satellitetoday.com/connectivity/2005/08/01/satellite-operators-target-interference-problem/. SUIRG database of 3,650 incidents: 62% human error, cross-pol and equipment. SUIRG president Robert Ames is quoted: “another 34 percent of interference incidents are caused by unknown sources, usually because the problems appear and disappear before the operators have a chance to document it.” Tags: OSR motivation.
80. **Coleman, M. (sIRG), Microwave Journal (14 Aug 2012).** “Tackling Satellite Interference.” https://www.microwavejournal.com/articles/18088-tackling-satellite-interference. Contains the quote “according to Intelsat, as much as 33 percent of interference is due to cross polarization leakage.” Intelsat also attributes 47% to unauthorized carriers. Includes Monics case studies. Tags: motivation.
81. **Kratos.** “Complete VSAT Interference Mitigation” (fact sheet). https://www.kratosspace.com/-/media/communications/pdf/fact-sheet---complete-vsat-interference-mitigation.pdf. Vendor material. It echoes IRG’s figure that VSATs “cause approximately 40% of all interference, and are responsible for 50% of downtime” (Martin Coleman, IRG, Dec 2014). The “50–70%” variant comes from Kratos marketing copy. Tags: motivation.
82. **Kratos (2026).** “Kratos KnownSpace Detects Sustained Iran Origin RF Interference Targeting GEO Communications.” https://www.kratosspace.com/sdatracker/kratos-knownspace-detects-sustained-iran-origin-rf-interference-targeting-geo-communications. Tags: unknown-class motivation.
83. **Rohde & Schwarz.** R&S GSACSM product page. https://www.rohde-schwarz.com/us/products/aerospace-defense-security/secure-communications-options/rs-gsacsm-communications-system-monitoring_63493-636288.html. Covers PCMA and under-carrier detection by carrier cancellation. rohde-schwarz
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Tags: [BASE].
84. **Integrasys.** VectorSat. https://www.integrasys-space.com/interference-detection. Tags: [BASE].
85. **Calian.** Decimator carrier-under-carrier detection. https://www.calian.com/advanced-technologies/space-satcom/decimator-carrier-under-carrier-detection/. Tags: [BASE].
86. **“A Novel and Blind Detection Scheme for Asymmetric PCMA System.”** IEEE, 2014. https://ieeexplore.ieee.org/document/6967240/. Paywalled. Tags: C2.
87. **“Information Theory Analysis of Blind Detection for PCMA Satellite Communication Systems.”** OA PDF (author upload, ResearchGate). Tags: C2 identifiability.
88. **“Cross-polarisation Discrimination-Induced Interference in Dual-Polarised High-Capacity Satellite Communication Systems.”** Landing: https://www.researchgate.net/publication/303477998. Tags: C1 (XPD statistics).
89. **“Impact of Cross-Polarization Interference on Dual Polarization...”** JPL IPN Progress Report 42-213. OA PDF: https://ipnpr.jpl.nasa.gov/progress_report/42-213/42-213A.pdf. Tags: C1.

G. Classical baselines and domain shift

90. **Gardner, W. A.** “Exploitation of Spectral Redundancy in Cyclostationary Signals.” IEEE SP Magazine 8(2), 1991. DOI: 10.1109/79.81007 (verify). Paywalled. Tags: [BASE], C2.
91. **Chen, T., Guestrin, C.** “XGBoost.” KDD 2016. OA: https://arxiv.org/abs/1603.02754. Tags: [BASE].
92. **“Approaching Domain Generalization with Embeddings for Robust Discrimination and Recognition of RF Communication Signals.”** OA PDF: https://arxiv.org/pdf/2510.23186. Tags: C3.
93. **“RF-Analyzer: Can Vision-Language Models Learn RF Understanding from Synthetic Data?”** OA: https://arxiv.org/html/2605.04676v1. Tags: C3 (sim-to-real).

Dataset and Tool Inventory

Name	URL	License	Contents	Usability
SnT Satellite On-board Interference	https://fnr-smartspace-project.uni.lu/satellite-on-board-interference-detection/	Free, **research-only, non-commercial**	4 FFT representations; 12,000 train / 3,000 val; 6 subband classes uni Ñ uni Ñ	Baseline reproduction of Daoud et al.; no payload effects
SnT NGSO to GSO's Users Interference	https://fnr-smartspace-project.uni.lu/ngso-to-gsos-users-interference/	Research-only	Time and spectrum magnitude (800 points); 11,500 interference-free training; 4,470 mixed test; QPSK/8PSK/16APSK test sets uni Ñ	Detection baseline (vs. TrID/Agon); not class-labelled by interference type
Fraunhofer GNSS Highway 1	https://ieee-dataport.org/documents/gnss-interference-spectrum-highway-dataset-1 (doi:10.21227/xpm9-tt28)	CC BY 4.0	Real highway snapshots, E1/E6	Real-interferer realism and OSR protocol template
Highway 2; LC Laboratory; Jammertest 2025	https://zenodo.org/records/21336653 · https://zenodo.org/records/21333335 · https://zenodo.org/records/21332689	CC BY-SA 4.0	Real and controlled GNSS interference; domain-shift splits	Unknown-class realism (pulsed, chirp, bursts)
TorchSig / Sig53 / WBSig53	https://torchsig.com (paper arXiv 2211.10335)	Open source (check repo license)	53-class narrowband; 550k wideband spectrogram samples arxiv Ñ semanticscholar Ñ	Pre-training, wideband detector baseline, waveform generation
OpenDPD (v1/v2)	https://github.com/lab-emi/OpenDPD	Open source (check repo)	Measured PA I/O (DPA_200MHz), PyTorch PA models themoonlight Ñ github Ñ	Amplifier-mismatch robustness; not a TWTA

Name	URL	License	Contents	Usability
DVB-S2X guidelines (TR 102 376-2)	dvb.org PDF above	Free	TWTA AM/AM–AM/PM curves, IMUX/OMUX, phase-noise mask dvb N̈	Simulator traceability
Chalmers RF WebLab	(not verified in this search)	—	Remote PA measurement	Possible TWTA-like measurements; verify access

Recommendations

Risks

1. **The "simulation-only" reviewer objection.** Every comparable GEO paper is simulation-based, but TAES or TMLCN reviewers will ask about real data. Mitigate by calibrating TWTA curves to the DVB-S2X tables, validating IM levels via NPR literature, testing amplifier-model mismatch (train Saleh, test Rapp, memory polynomial or an OpenDPD-learned model), and, if possible, obtaining a handful of operator spectrum-analyzer traces for qualitative checks.
2. **Shortcut learning on the plan mask.** The mask can trivially reveal "unauthorized carrier" (energy outside the mask). Include a plan-perturbation test with stale or erroneous plans, and an ablation with the mask present but carriers moved.
3. **Carrier-under-carrier is not observable in the spectrogram.** Unless the cyclic branch is present, expect near-chance performance at low C/I. Treat this as an identifiability *result*, not a failure.
4. **Choice of unknowns.** Wideband noise versus overdrive-IM regrowth, and pulsed versus intermittent bursts, are near-duplicates. Report per-unknown AUROC and separate "near" from "far" OOD.
5. **Overclaiming "first."** Agon, the Vázquez/Eutelsat work and the open-set jamming papers partially overlap. Use precise wording such as "first to jointly model…".

Refinements to contributions

- C1: add randomized payload parameters (TWTA family, IMUX/OMUX order, XPD distribution) as domain randomization, and publish the generator with a fixed benchmark split and dataset card, following the S-ICDF and Fraunhofer releases.
- C2: write the identifiability analysis as a table of class × observable (co-pol PSD, cross-pol PSD, plan, IBO response, cyclic features). Back it with the ablations, and add FiLM conditioning as an alternative to the mask channel.
- C3: report the payload gap as a function of IBO distance from training, not as a single number.

Additional baselines and metrics

- Baselines: EDSC (signal-cancellation ED); a TrID/CAE reconstruction detector; the WBSig53 pre-trained detector; ARPL and OpenMax; a hybrid DL-plus-expert classifier (arXiv 2302.03749).
- Metrics: OSCR, FPR@95TPR, per-unknown AUROC, expected calibration error, Pd at fixed Pfa versus C/I and versus IBO, and inference latency per spectrum sweep (operators scan up to "200 carriers per second," per the Integrasys vendor claim).

Top-15 prioritized reading list

1. Saifaldawla et al., TMLCN 2024 (GenAI NGSO detection)
2. Yang et al., Agon, arXiv 2026
3. Daoud et al., ICASSP 2023
4. Casini, De Gaudenzi & Ginesi, IJSCN 2004
5. DVB-S2X Implementation Guidelines (TR 102 376-2)
6. Vaze et al., ICLR 2022
7. Liu et al., energy OOD, NeurIPS 2020
8. Lee et al., Mahalanobis, NeurIPS 2018
9. Dhamija et al., OSCR, NeurIPS 2018
10. Heublein et al., ION GNSS+ 2024 (real-world discrepancies)
11. Vázquez et al., Commun. Mag. 2021, plus the 2019 newsletter
12. Boegner et al., WBSig53, 2022
13. Efficient OSR for interference signals (Appl. Sci. 2022)
14. Beidas, multicarrier IM analysis (IEEE TCOM)
15. Tekbıyık et al., SCF-CNN (arXiv 2003.08359)

Caveats

- **Search coverage.** The search budget was limited. CNKI-indexed Chinese literature and full IEEE Xplore texts were only partly checked, so a closer GEO paper could exist.
- **Unconfirmed details.** Not verified from full text: whether the Commun. Mag. 2021 interference data are Eutelsat transponder spectra (check the Zenodo PDF); the Mehr & Dovis DOI; the Chalmers RF WebLab license. DOIs marked “(verify)” were recalled from memory. CA-CFAR’s origin (Finn & Johnson, RCA Review 29, 1968) is print-only.
- **Unrefereed sources.** Agon (2026) and several 2026 arXiv items are preprints and may change.